

Vector Spaces

Real Vector Space:

The vector spaces with real scalars are called *real vector spaces*.

Complex Vector Space:

The vector spaces with complex scalars are called *complex vector spaces*.

Note:

For now, we will be concerned exclusively with real vector spaces.

Real Vector Spaces:

The following defines the notion of a **vector space** V , where K is the **field** of scalars.

Definition:

Let V be a non-empty set of objects, on which **two operations** are defined:

i. **Vector Addition:**

ii. **Scalar Multiplication:**

then V is called a **vector space** over the field K , if the following axioms hold, for any $\vec{u}, \vec{v}, \vec{w} \in V$:

- 1) Closure Law of Addition: If $\vec{u}, \vec{v} \in V$, then $\vec{u} + \vec{v} \in V$. (V is closed under addition)
- 2) Associative Law of Addition: $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$
- 3) Additive Identity: There is a vector in V , called the **zero vector**, denoted by $\vec{0}$ such that, for any $\vec{u} \in V$

$$\vec{u} + \vec{0} = \vec{0} + \vec{u} = \vec{u}$$

- 4) Additive Inverse: For each $\vec{u} \in V$, there is a vector in V , denoted by $-\vec{u}$, called a **negative of $\vec{u}$** , such that

$$\vec{u} + (-\vec{u}) = (-\vec{u}) + \vec{u} = 0$$

5) **Commutative Law of Addition:** $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

6) **Scalar Multiplication:** If k is any scalar element of the field K and $\vec{u}$ is any vector in V , then $k\vec{u}$ is in V . (**Closed under Scalar Multiplication**)

7) For any scalar $k \in K$, we have $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$

8) For any scalars $a, b \in K$, we have $(k + m)\vec{u} = k\vec{u} + m\vec{u}$

9) For any scalars $a, b \in K$, we have $k(m\vec{u}) = (km)\vec{u}$

10) **Multiplicative Identity:** For the unit scalar $1 \in K$, we have $1\vec{u} = \vec{u}$

Example # 1: Let

$$V = R^2 = \{(x, y); x, y \in R\}$$

Prove that V is a **vector space** under the **usual (standard) operations of addition and scalar multiplication** defined by:

$$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) \text{ --- (1)}$$

$$k\vec{u} = k(u_1, u_2) = (ku_1, ku_2) \text{ --- (2)}$$

Solution:

1. **Closure Law of Addition:** V is closed under addition [as defined in equation (1)]

2. **Associative Law of Addition:** Let

$$\vec{u} = (u_1, u_2), \quad \vec{v} = (v_1, v_2), \quad \vec{w} = (w_1, w_2)$$

then

$$(\vec{u} + \vec{v}) + \vec{w} = ((u_1, u_2) + (v_1, v_2)) + (w_1, w_2)$$

$$(\vec{u} + \vec{v}) + \vec{w} = (u_1 + v_1, u_2 + v_2) + (w_1, w_2)$$

$$(\vec{u} + \vec{v}) + \vec{w} = (u_1 + v_1 + w_1, u_2 + v_2 + w_2)$$

$$(\vec{u} + \vec{v}) + \vec{w} = (u_1 + (v_1 + w_1), u_2 + (v_2 + w_2))$$

$$(\vec{u} + \vec{v}) + \vec{w} = (u_1, u_2) + (v_1 + w_1, v_2 + w_2)$$

$$(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$$

3. **Additive Identity:** Let $\vec{u} = (u_1, u_2)$, $\vec{0} = (0,0)$

$$\vec{u} + \vec{0} = (u_1, u_2) + (0,0) = (u_1, u_2) = \vec{u}$$

4. **Additive Inverse:** Let $\vec{u} = (u_1, u_2)$, then there exist $-\vec{u} = (-u_1, -u_2)$,

$$\vec{u} + (-\vec{u}) = (u_1 + (-u_1), u_2 + (-u_2)) = (u_1 - u_1, u_2 - u_2) = (0,0) = \vec{0}$$

5. **Commutative Law of Addition:** Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$, then

$$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2)$$

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2)$$

$$\vec{u} + \vec{v} = (v_1 + u_1, v_2 + u_2)$$

$$\vec{u} + \vec{v} = (v_1, v_2) + (u_1, u_2)$$

$$\vec{u} + \vec{v} = \vec{v} + \vec{u}$$

6. **Scalar Multiplication:** V is **closed** under **scalar multiplication**. [as defined in equation (2)]

$$7. \quad k(\vec{u} + \vec{v}) = k((u_1, u_2) + (v_1, v_2))$$

$$k(\vec{u} + \vec{v}) = k(u_1 + v_1, u_2 + v_2)$$

$$k(\vec{u} + \vec{v}) = (ku_1 + kv_1, ku_2 + kv_2)$$

$$k(\vec{u} + \vec{v}) = (ku_1, ku_2) + (kv_1, kv_2)$$

$$k(\vec{u} + \vec{v}) = k(u_1, u_2) + k(v_1, v_2)$$

$$k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$$

$$8. \quad (k + m)\vec{u} = (k + m)(u_1, u_2)$$

$$(k + m)\vec{u} = (ku_1 + mu_1, ku_2 + mu_2)$$

$$(k + m)\vec{u} = (ku_1, ku_2) + (mu_1, mu_2)$$

$$(k + m)\vec{u} = k(u_1, u_2) + m(u_1, u_2)$$

$$(k + m)\vec{u} = k\vec{u} + m\vec{u}$$

$$9. \quad k(m \vec{u}) = k(m(u_1, u_2)) = (k m u_1, k m u_2)$$

$$k(m \vec{u}) = km(u_1, u_2) = k m (\vec{u})$$

$$10. \quad 1\vec{u} = 1(u_1, u_2) = (u_1, u_2) = \vec{u}$$

Since the set V satisfies all the properties, so V is a **vector space**.

Example # 2: Let $V = R^3 = \{(x, y, z) \mid x, y, z \in R\}$, prove that V is a vector space under the **usual (standard) operations of addition and scalar multiplication** defined by:

$$\vec{u} + \vec{v} = (u_1, u_2, u_3) + (v_1, v_2, v_3) = (u_1 + v_1, u_2 + v_2, u_3 + v_3) \text{ --- (1)}$$

$$k\vec{u} = k(u_1, u_2, u_3) = (ku_1, ku_2, ku_3) \text{ --- (2)}$$

Solution: Students will do themselves

Example # 3: Let $V = R^2$, under the **usual (standard) operations of addition** defined by:

$$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) \text{ --- (1)}$$

And if k is any scalar element of **field** K , then define

$$k\vec{u} = k(u_1, u_2) = (ku_1, 0) \text{ --- (2)}$$

Check whether V is vector space or not?

Solution: All properties of addition are satisfied. (**Check it by yourself**)

Let us check the properties of scalar multiplication.

$$6. \text{ Let } \vec{u} = (u_1, u_2) \text{ in } V, \text{ then } k\vec{u} = k(u_1, u_2) = (ku_1, 0) \in V.$$

$$7. \text{ Let } \vec{u} = (u_1, u_2), \quad \vec{v} = (v_1, v_2)$$

$$k(\vec{u} + \vec{v}) = k((u_1, u_2) + (v_1, v_2))$$

$$k(\vec{u} + \vec{v}) = k(u_1 + v_1, u_2 + v_2)$$

$$k(\vec{u} + \vec{v}) = (k u_1 + k v_1, 0)$$

$$k(\vec{u} + \vec{v}) = (k u_1, 0) + (k v_1, 0)$$

$$k(\vec{u} + \vec{v}) = k(u_1, 0) + k(v_1, 0)$$

Since $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$ and $\vec{u} \neq (u_1, 0)$, $\vec{v} \neq (v_1, 0)$, therefore,

$$k(\vec{u} + \vec{v}) \neq k \vec{u} + k \vec{v}$$

As the 7th property does not satisfy. So, it is **not** a vector space.

Example 4:

Check whether V is vector space or not?

V = Set of all pairs of real numbers of the form $(x, 0)$. i.e., $\{(x, 0); x \in R\}$

with the **standard operations** on R^2 .

$$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) \dots (1)$$

$$k\vec{u} = k(u_1, u_2) = (ku_1, ku_2) \dots (2)$$

Solution:

1. **Closure Law of Addition:** Let $\vec{u} = (u_1, 0)$, $\vec{v} = (v_1, 0) \in V$, then

$$(\vec{u} + \vec{v}) = (u_1 + v_1, 0) \in V$$

So, V is closed under addition.

2. **Associative Law of Addition:**

Let $\vec{u} = (u_1, 0)$, $\vec{v} = (v_1, 0)$, $\vec{w} = (w_1, 0) \in V$, then

$$\vec{u} + (\vec{v} + \vec{w}) = (u_1, 0) + ((v_1, 0) + (w_1, 0))$$

$$\vec{u} + (\vec{v} + \vec{w}) = (u_1, 0) + (v_1 + w_1, 0)$$

$$\vec{u} + (\vec{v} + \vec{w}) = (u_1 + v_1 + w_1, 0)$$

$$\vec{u} + (\vec{v} + \vec{w}) = (u_1 + v_1, 0) + (w_1, 0)$$

$$\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$$

3. **Additive Identity:** Let $\vec{u} = (u_1, 0)$, then

$$\vec{u} + \vec{0} = (u_1, 0) + (0, 0) = (u_1, 0) = \vec{u}$$

4. **Additive Inverse:** Let $\vec{u} = (u_1, 0)$, then

$$\begin{aligned}\vec{u} + (-\vec{u}) &= (u_1, 0) + (-u_1, 0) \\ &= (u_1 - u_1, 0) = (0, 0) = \vec{0}\end{aligned}$$

5. **Commutative law of Addition:** Let $\vec{u} = (u_1, 0)$, $\vec{v} = (v_1, 0) \in V$, then

$$\begin{aligned}\vec{u} + \vec{v} &= (u_1, 0) + (v_1, 0) \\ \vec{u} + \vec{v} &= (u_1 + v_1, 0) \\ \vec{u} + \vec{v} &= (v_1 + u_1, 0) \\ \vec{u} + \vec{v} &= (v_1, 0) + (u_1, 0) \\ \vec{u} + \vec{v} &= \vec{v} + \vec{u}\end{aligned}$$

6. **Scalar Multiplication:** Let $\vec{u} = (u_1, 0) \in V$, then

$$k\vec{u} = (ku_1, k0) = (ku_1, 0) \in V$$

$$\begin{aligned}7. \quad k(\vec{u} + \vec{v}) &= k((u_1, 0) + (v_1, 0)) = k(u_1 + v_1, 0) = (ku_1 + kv_1, 0) \\ &= (ku_1, 0) + (kv_1, 0) = k(u_1, 0) + k(v_1, 0) \\ &= (k\vec{u} + k\vec{v})\end{aligned}$$

$$\begin{aligned}8. \quad (k + m)\vec{u} &= (k + m)(u_1, 0) \\ &= ((k + m)u_1, 0) = (ku_1 + mu_1, 0) \\ &= (ku_1, 0) + (mu_1, 0) \\ &= k(u_1, 0) + m(u_1, 0) = k\vec{u} + m\vec{u}\end{aligned}$$

$$\begin{aligned}9. \quad k(m\vec{u}) &= k(mu_1, 0) = (kmu_1, 0) \\ &= km(u_1, 0) = (km)\vec{u}\end{aligned}$$

$$10. \quad 1\vec{u} = 1(u_1, 0) = (u_1, 0) = \vec{u}$$

Since all properties of vectors space are satisfied by the set V . Therefore, V is a vector space.

Example # 5: Check whether V is a vector space or not.

V = Set of all pairs of real numbers of the form (x, y) , where $x \geq 0$, i.e.

$$V = \{(x, y); x \geq 0, y \in \mathbb{R}\}$$

with **standard operations** on $\mathbb{R}^2$.

Solution:

As

$$V = \{(x, y); x \geq 0, y \in \mathbb{R}\}$$

1. Let $\vec{u} = (u_1, u_2), \quad \vec{v} = (v_1, v_2) \in V$

$$(\vec{u} + \vec{v}) = (u_1 + v_1, u_2 + v_2) \in V$$

Because $\vec{u}_1 + \vec{v}_1 \geq 0$. So, V is closed under addition.

2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (Easy to verify)

3. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$ (Easy to verify)

4. Let $\vec{u} = (u_1, u_2), \quad \vec{u} + \vec{0} = (u_1, u_2) + (0, 0) = (u_1, u_2) = \vec{u}$

5. Let $\vec{u} = (u_1, u_2)$, Then there doesn't exist $-\vec{u} = (-u_1, -u_2)$ because $\vec{u}_1$ should be positive.

Hence, 5th property fails, So, V is not vector space.

Example 6: Show that the set of all pairs of real numbers of the form $(x, 1)$ with the operations

$$(x, 1) + (x', 1) = (x + x', 1)$$

and

$$k(x, 1) = (k^2x, 1)$$

is not a vector space.

Example # 7: Determine whether the set of all triples of real numbers **with standard vector addition** but with **scalar multiplication defined by**

$$k(x, y, z) = (k^2x, k^2y, k^2z)$$

is a vector space or not?

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•
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Axiom 8 fails.

Example # 8: Determine whether the set of all pairs of real numbers of the form $(1, x)$ with the operations

$$(1, y) + (1, y') = (1, y + y')$$

and

$$k(1, y) = (1, ky)$$

is a vector space or not?

Example # 9: Determine whether V is a vector space or not.

V = the set of all triples of the form (x, y, z) with the operations

$$(x, y, z) + (x', y', z') = (x + x', y + y', z + z')$$

&

$$k(x, y, z) = (kx, y, z)$$

Example # 10: Let V be the set of all 2×2 matrices with **real entries** and take the vector space operations on V to be **usual operations of matrix addition and scalar multiplication** i.e.

$$\vec{u} + \vec{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11} + v_{11} & u_{12} + v_{12} \\ u_{21} + v_{21} & u_{22} + v_{22} \end{bmatrix} \text{ --- (1)}$$

$$k\vec{u} = k \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix} \text{ --- (2)}$$

Determine whether V is a vector space or not?

Solution:

1. From equation (1), it is clear that V is **closed** under addition.

2. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$

$$3. \vec{u} + 0 = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \vec{u}$$

$$\begin{aligned} 4. \vec{u} + (-\vec{u}) &= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} -u_{11} & -u_{12} \\ -u_{21} & -u_{22} \end{bmatrix} \\ &= \begin{bmatrix} u_{11} + (-u_{11}) & u_{12} + (-u_{12}) \\ u_{21} + (-u_{21}) & u_{22} + (-u_{22}) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \vec{0} \end{aligned}$$

$$\begin{aligned} 5. \quad \vec{u} + \vec{v} &= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \\ &= \begin{bmatrix} u_{11} + v_{11} & u_{12} + v_{12} \\ u_{21} + v_{21} & u_{22} + v_{22} \end{bmatrix} \\ &= \begin{bmatrix} v_{11} + u_{11} & v_{12} + u_{12} \\ v_{21} + u_{21} & v_{22} + u_{22} \end{bmatrix} \\ &= \vec{v} + \vec{u} \end{aligned}$$

Similarly, you can prove all the properties of scalar multiplication.

(Prove it by yourself).

Since all properties of vector space with **usual operations of matrix addition and scalar multiplication** satisfied by the given set V , therefore, V is a vector space.

Example # 11: Let $V = \mathbf{R}^n$ and define operations on V to be the usual operations of addition given by

$$\vec{u} + \vec{v} = (u_1, u_2, u_3, \dots, u_n) + (v_1, v_2, v_3, \dots, v_n) = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

and scalar multiplication given by

$$k\vec{u} = k(u_1, u_2, u_3, \dots, u_n) = (ku_1, ku_2, ku_3, \dots, ku_n)$$

Show that V is a vector space.

Example # 12: Let V be the set of polynomials of the form

$$P(t) = a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0.$$

Determine whether V is a vector space under the **usual operations of addition and scalar multiplication** or not?

THEOREM:

Let V be a vector space, $\mathbf{u}$ be a vector in V , and k be a scalar; then:

- a) $0 \mathbf{u} = \mathbf{0}$
- b) $k \mathbf{0} = \mathbf{0}$
- c) $(-1)\mathbf{u} = -\mathbf{u}$
- d) If $k \mathbf{u} = \mathbf{0}$, then $k = 0$ or $\mathbf{u} = \mathbf{0}$.

Practice Questions for Students

Q # 1: Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\vec{\mathbf{u}} = (\mathbf{u}_1, \mathbf{u}_2)$ and $\vec{\mathbf{v}} = (\mathbf{v}_1, \mathbf{v}_2)$:

$$\vec{\mathbf{u}} + \vec{\mathbf{v}} = (\mathbf{u}_1, \mathbf{u}_2) + (\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{v}_1, \mathbf{u}_2 + \mathbf{v}_2) \text{ --- (1)}$$

$$k\vec{\mathbf{u}} = k(\mathbf{u}_1, \mathbf{u}_2) = (\mathbf{0}, k\mathbf{u}_2) \text{ --- (2)}$$

- a) Compute $\vec{\mathbf{u}} + \vec{\mathbf{v}}$ and $k\vec{\mathbf{u}}$ for $\vec{\mathbf{u}} = (-1, 2)$, $\vec{\mathbf{v}} = (3, 4)$ and $k = 3$.
- b) Since addition of V is standard operations on $\mathbf{R}^2$, certain vector space axioms hold for V because they are known to hold for $\mathbf{R}^2$. Which axioms are they?
- c) If $\vec{\mathbf{u}} = (-1, 2)$, then find the **addition inverse** of $\vec{\mathbf{u}}$.
- d) Show that **axiom 7, 8, & 9** hold, where $m = 2$.
- e) Show that **axiom 10 fails** and hence V is not a vector space under the given operations.

Answer: Do yourself

- (a) $\vec{\mathbf{u}} + \vec{\mathbf{v}} = (2, 6)$ and $k\vec{\mathbf{u}} = (\mathbf{0}, 6)$.
- c) **Axiom 1-5**

Q# 2: Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\vec{\mathbf{u}} = (\mathbf{u}_1, \mathbf{u}_2)$ and $\vec{\mathbf{v}} = (\mathbf{v}_1, \mathbf{v}_2)$:

$$\vec{\mathbf{u}} + \vec{\mathbf{v}} = (\mathbf{u}_1, \mathbf{u}_2) + (\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{v}_1 + 1, \mathbf{u}_2 + \mathbf{v}_2 + 1) \text{ --- (1)}$$

$$k\vec{\mathbf{u}} = k(\mathbf{u}_1, \mathbf{u}_2) = (k\mathbf{u}_1, k\mathbf{u}_2) \text{ --- (2)}$$

- a) Compute $\vec{\mathbf{u}} + \vec{\mathbf{v}}$ and $k\vec{\mathbf{u}}$ for $\vec{\mathbf{u}} = (\mathbf{0}, 4)$, $\vec{\mathbf{v}} = (1, -3)$ and $k = 2$.
- b) If $\vec{\mathbf{u}} = (-1, 2)$, then find the **addition inverse** of $\vec{\mathbf{u}}$.
- c) Show that **axiom 5** holds by producing an ordered pair $-\vec{\mathbf{u}}$ such that $\vec{\mathbf{u}} + (-\vec{\mathbf{u}}) = \mathbf{0}$ for $\vec{\mathbf{u}} = (\mathbf{u}_1, \mathbf{u}_2)$
- d) Find two vector space **axioms that fail** to hold.